

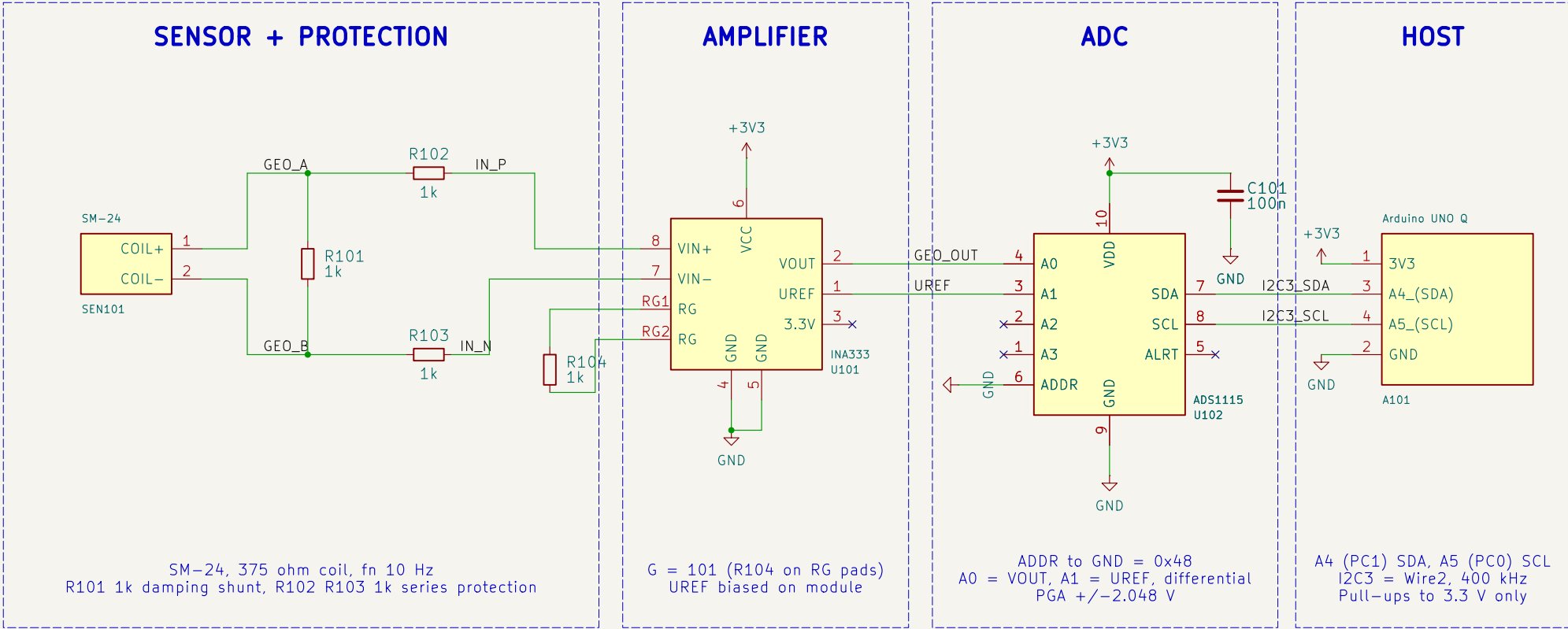
|   |                                                                                                                                          |                                                                      |                                                                         |                                                                |   |   |   |   |  |  |  |
|---|------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------|-------------------------------------------------------------------------|----------------------------------------------------------------|---|---|---|---|--|--|--|
| 1 | 2                                                                                                                                        | 3                                                                    | 4                                                                       | 5                                                              | 6 | 7 | 8 |   |  |  |  |
| A | <div>EleTect X – SHEET INDEX</div> <div>Sheets 1 geophone, 2 LED and IR, 3 sound and LoRa, 4 power: one project, one Arduino UNO Q</div> |                                                                      |                                                                         |                                                                |   |   |   |   |  |  |  |
| B | <div>Geophone</div> <div>File: geophone.kicad_sch</div> <div>1</div>                                                                     | <div>LED and IR</div> <div>File: led_ir.kicad_sch</div> <div>2</div> | <div>Sound and LoRa</div> <div>File: audio.kicad_sch</div> <div>3</div> | <div>Power</div> <div>File: power.kicad_sch</div> <div>4</div> |   |   |   |   |  |  |  |
| C |                                                                                                                                          |                                                                      |                                                                         |                                                                |   |   |   |   |  |  |  |
| D |                                                                                                                                          |                                                                      |                                                                         |                                                                |   |   |   |   |  |  |  |
| E |                                                                                                                                          |                                                                      |                                                                         |                                                                |   |   |   |   |  |  |  |
| F | 1                                                                                                                                        | 2                                                                    | 3                                                                       | 4                                                              | 5 | 6 | 7 | 8 |  |  |  |

|                                         |                  |        |
|-----------------------------------------|------------------|--------|
| EleTect X   Designed by Abhinav Krishna |                  |        |
| Sheet: /                                |                  |        |
| File: eletect_x.kicad_sch               |                  |        |
| Title: EleTect X – Sheet Index          |                  |        |
| Size: A3                                | Date: 2026-10-04 | Rev: A |
| KiCad E.D.A. 10.0.5                     | Id: 5/5          |        |

|                                         |                  |        |
|-----------------------------------------|------------------|--------|
| EleTect X   Designed by Abhinav Krishna |                  |        |
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| File: eletect_x.kicad_sch               |                  |        |
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# EleTect X – GEOPHONE ANALOG FRONT-END BOARD

SM-24 geophone > INA333 instrumentation amplifier > ADS1115 16-bit ADC > Arduino UNO Q (I2C3)



Sheet 1 of 4 | designators 1xx  
Drawn by Abhinav Krishna – EleTect X  
3V3 supply. R101–R104 1k, C101 100n  
SM-24 > INA333 (G=101) > ADS1115 (0x48) > UNO Q  
**EleTect X | Designed by Abhinav Krishna**

Sheet: /Geophone/  
File: geophone.kicad\_sch

**Title: Geophone Analog Front-End Board**

Size: A4 Date: 2026-10-04

KiCad E.D.A. 10.0.5

Rev: A

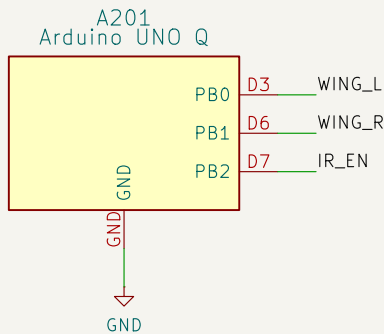
Id: 1/5

# EleTect X – LED WING + IR ILLUMINATOR MOSFET DRIVERS

Arduino UNO Q GPIO > gate resistor > IRLZ44N low-side switch > LED wings and 940 nm IR board

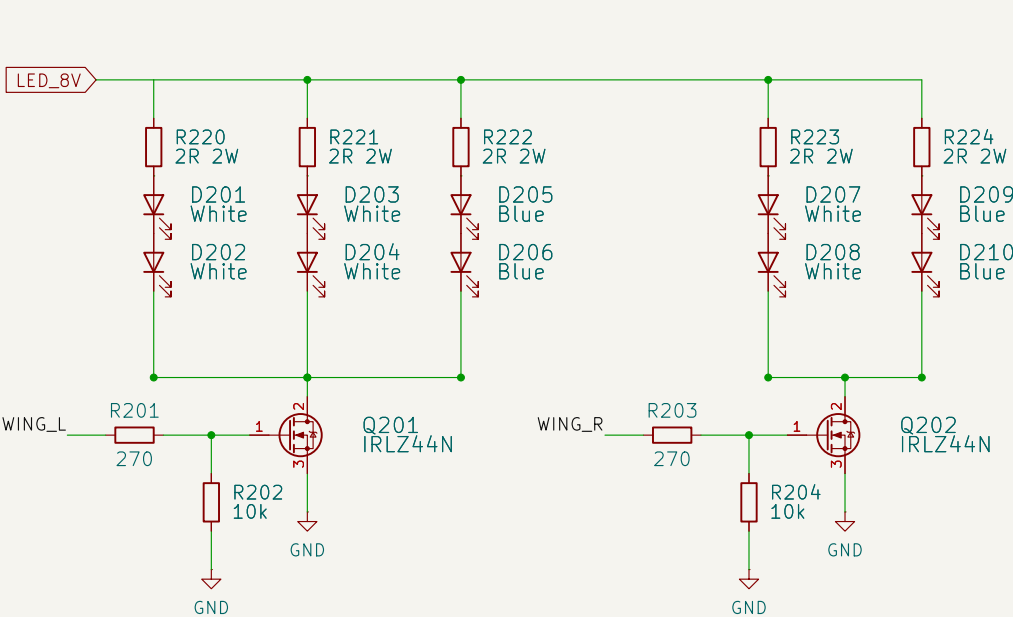
LED\_8V is the 8 V LED buck output and IR\_VBAT the fused battery bus, both from the power sheet; battery, fuses and switch live there

## HOST (Arduino UNO Q)



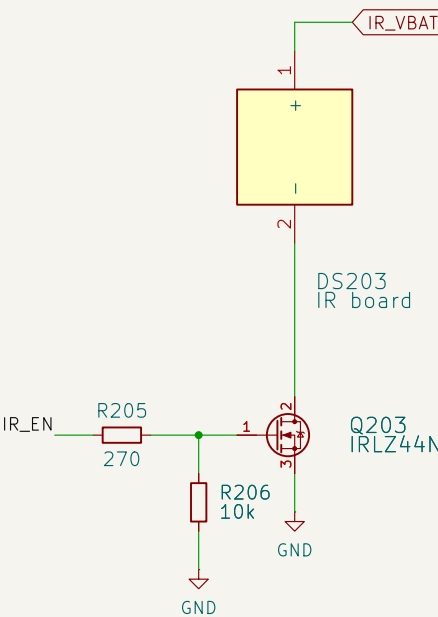
GPIO outputs, 3.3 V logic  
D3/PB0 is not 5 V tolerant:  
no pull-up above 3.3 V on it

## LED WINGS (white + blue)



Fed from the 8 V LED buck on the power sheet; two LEDs in series per string (6.6 V)  
2R 2 W sets about 700 mA per string at Vf 3.3 V; trim on the bench  
270R limits gate inrush, 10k pulldown holds the MOSFET off at boot

## IR ILLUMINATOR (940 nm)



Sheet 2 of 4 | designators 2xx  
Drawn by Abhinav Krishna – EleTect X  
Rails LED\_8V (8 V buck) and IR\_VBAT (fused battery bus) come from the power sheet  
UNO Q GPIO > 270R > IRLZ44N low-side switch > load; 10k gate pulldown

**EleTect X | Designed by Abhinav Krishna**

Sheet: /LED and IR/  
File: led\_ir.kicad\_sch

**Title: LED Wing + IR Illuminator MOSFET Drivers**

Size: User Date: 2026-10-04  
KiCad E.D.A. 10.0.5

Rev: A  
Id: 2/5

EleTect X – DFPLAYER PRO + HORN AMP + SHARED UART

UNO Q USART1 > 74HC4053 > DFPlayer PRO / LoRa-E5; DFPlayer DACL > XH-M543 amp > Ahuja SUH-15 horn

Horn amp runs from the raw 12.8 V battery bus; DFPlayer PRO and LoRa-E5 run from the UNO Q 5 V pin

**HOST (UNO Q)**

A301 Arduino UNO Q

5V, 3V3

D0/RX, D1/TX, D4, D11

MCU\_RX, MCU\_TX, UART\_SEL, AMP\_EN

GND

D0 / D1 = USART1 (3.3 V pins)  
D4 select: LOW = DFPlayer PRO, HIGH = LoRa-E5  
D11 gate HIGH = amp powered

**SHARED UART (74HC4053 + DFPlayer PRO + LoRa-E5)**

U301 74HC4053

MCU\_TX, MCU\_RX, UART\_SEL

R305 10k

3V3, GND

X0, Y0, X1, Y1

U302 DFPlayer PRO

VIN, DACL, DACR, KEY, PLAY, L+, L-, R+, R-

5V, 3V3, GND

R301 10k, R306 1k

U303 Grove LoRa-E5

RX, TX, VCC, GND

3V3, GND

R302 10k, R307 1k

U304 XH-M543 amp

VCC, INL, GND, OUT+, OUT-

LS301 SUH-15

74HC4053 runs at 3.3 V; the horn always wins the shared port (ADR 0029)  
DFPlayer PRO 115200 baud, LoRa-E5 9600 baud: each switch re-opens Serial1  
10k pull-ups hold the deselected RX line idle high  
R305 pulls D4 low at reset: DFPlayer PRO selected by default  
1k series R306 / R307 limit clamp current if a module TX swings above 3.3 V  
Unused 4053 pins (C, Inh, VEE, Z, Z0, Z1) are tied to GND

**HORN (XH-M543 amp + Ahuja SUH-15)**

+BATBUS

F302 3A

PWR\_FLAG

D301 P6KE18CA

C306 470u 35V

U304 XH-M543 amp

VCC, INL, GND, OUT+, OUT-

LS301 SUH-15

PWR\_FLAG

AMP\_EN

R303 270

Q304 IRLZ44N

R304 10k

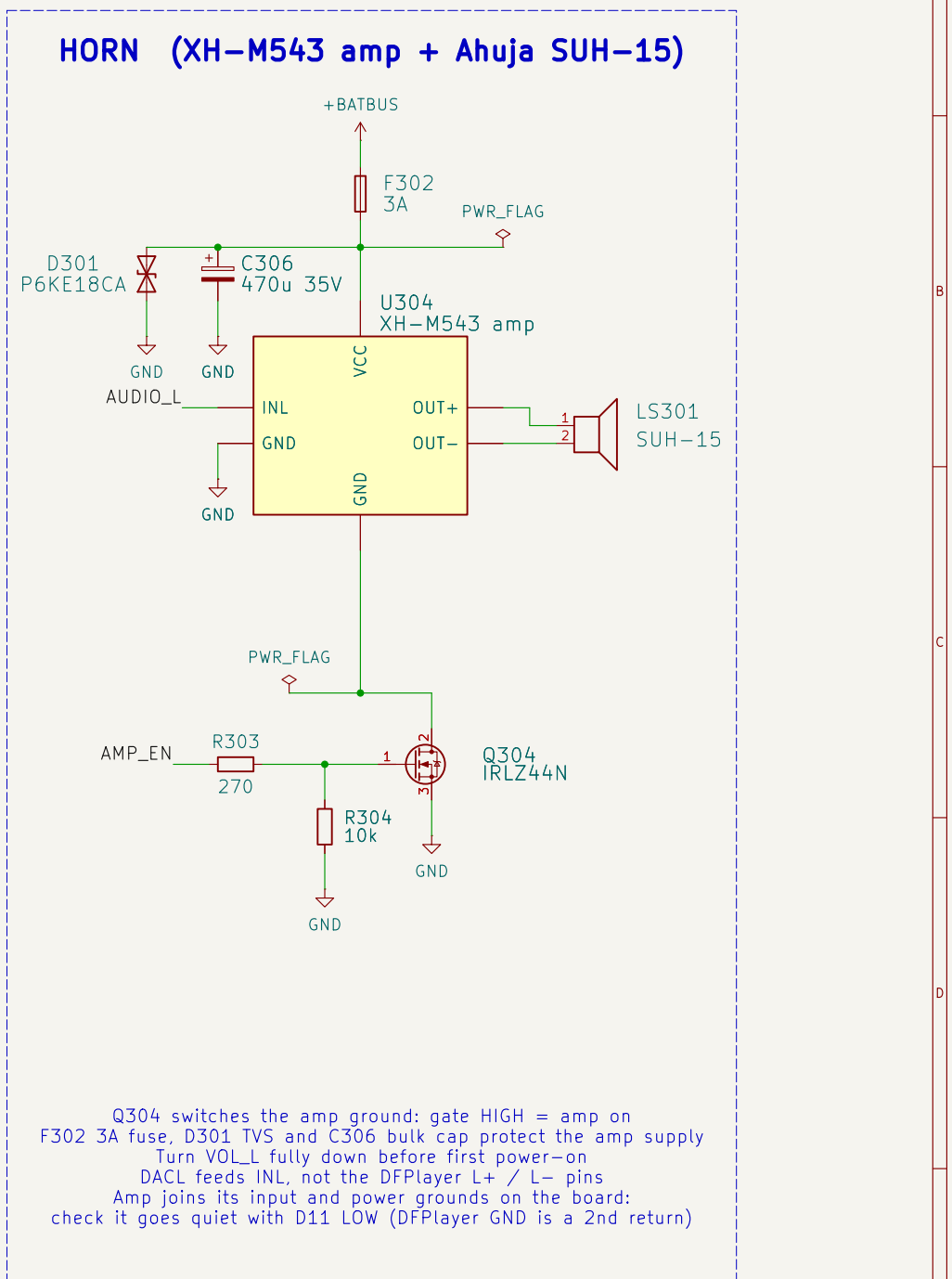
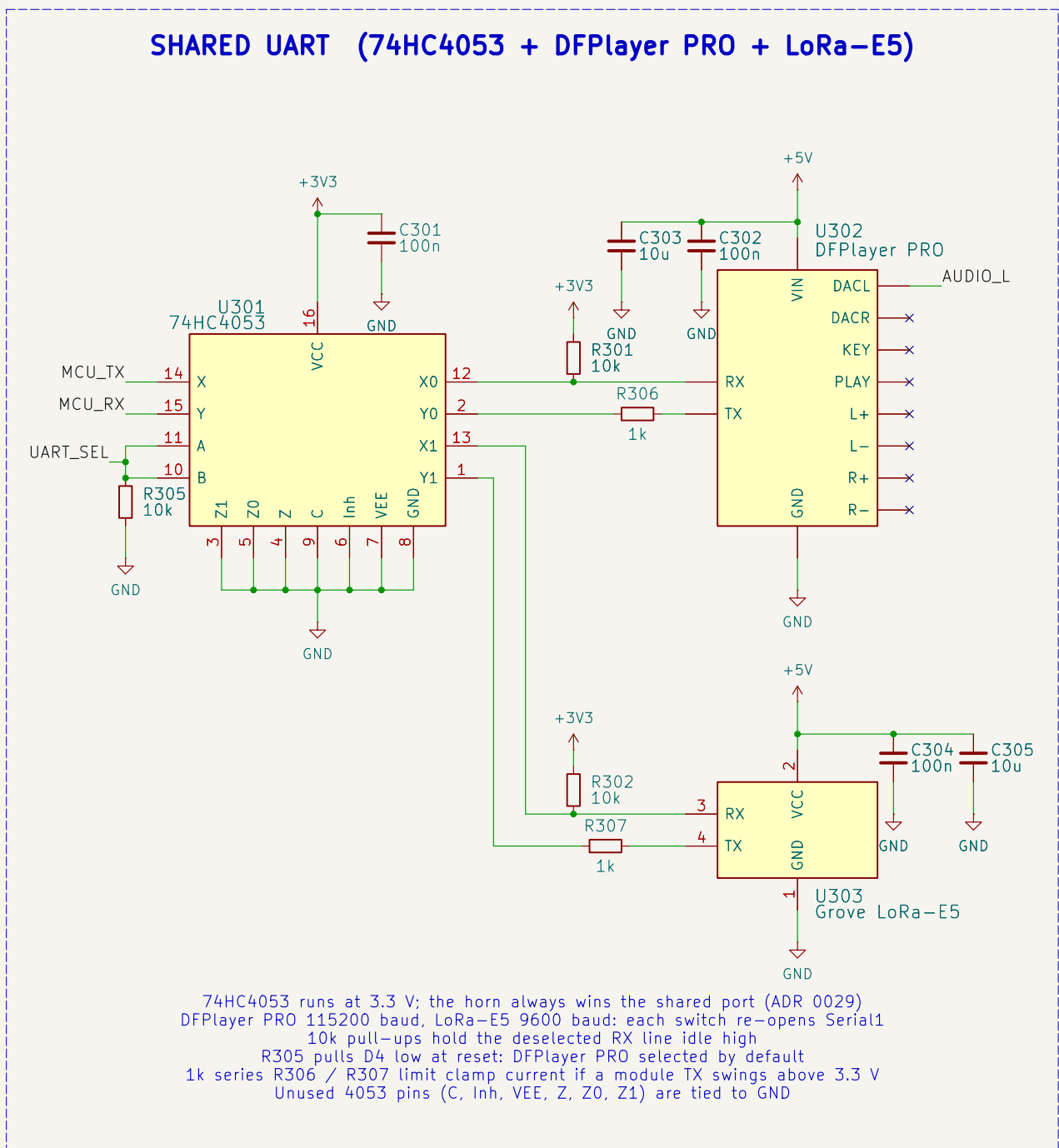
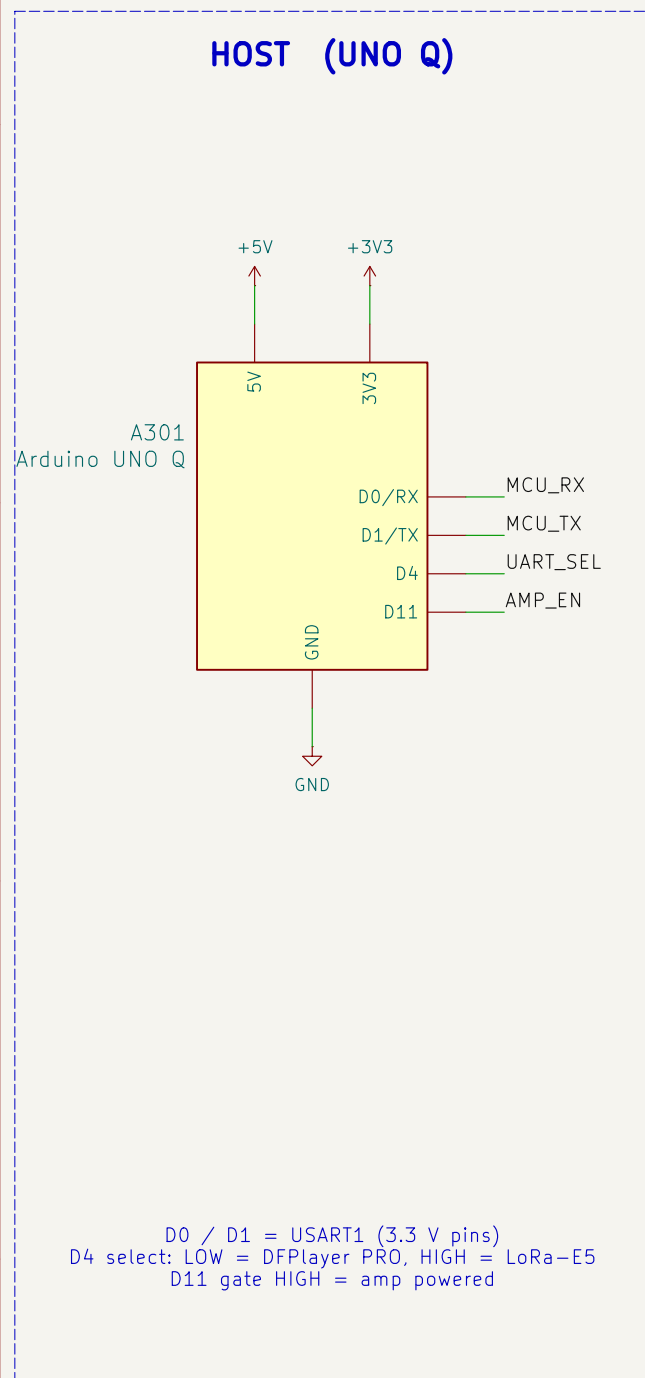
GND

Q304 switches the amp ground: gate HIGH = amp on  
F302 3A fuse, D301 TVS and C306 bulk cap protect the amp supply  
Turn VOL\_L fully down before first power-on  
DACL feeds INL, not the DFPlayer L+ / L- pins  
Amp joins its input and power grounds on the board:  
check it goes quiet with D11 LOW (DFPlayer GND is a 2nd return)

Sheet 3 of 4 | designators 3xx  
Drawn by Abhinav Krishna – EleTect X  
DACL > XH-M543 INL > Ahuja SUH-15; amp GND via Q304 (D11)  
USART1 D0/D1 shared via 74HC4053, select D4 (ADR 0029)  
**EleTect X | Designed by Abhinav Krishna**  
Sheet: /Sound and LoRa/  
File: audio.kicad\_sch  
**Title: DFPlayer PRO + Horn + Shared UART**  
Size: A3 | Date: 2026-10-04 | Rev: A  
KiCad E.D.A. 10.0.5 | Id: 3/5

UNO Q USART1 > 74HC4053 > DFPlayer PRO / LoRa-E5; DFPlayer DACL > XH-M543 amp > Ahuja SUH-15 horn

Horn amp runs from the raw 12.8 V battery bus; DFPlayer PRO and LoRa-E5 run from the UNO Q 5 V pin



|                                                           |                  |         |
|-----------------------------------------------------------|------------------|---------|
| Sheet 3 of 4   designators 3xx                            |                  |         |
| Drawn by Abhinav Krishna – EleTeC X                       |                  |         |
| DACL > XH-M543 INL > Ahuja SUH-15; amp GND via Q304 (D11) |                  |         |
| USART1 D0/D1 shared via 74HC4053, select D4 (ADR 0029)    |                  |         |
| <b>EleTeC X   Designed by Abhinav Krishna</b>             |                  |         |
| Sheet: /Sound and LoRa/                                   |                  |         |
| File: audio.kicad_sch                                     |                  |         |
| <b>Title: DFPlayer PRO + Horn + Shared UART</b>           |                  |         |
| Size: A3                                                  | Date: 2026-10-04 | Rev: A  |
| KiCad E.D.A. 10.0.5                                       |                  | Id: 3/5 |

# EleTect X – POWER: SOLAR, BATTERY, BUCKS

20 W panel > P6KE24CA TVS > SB5100 > MPPT controller (LiFePO4 mode) > 4S LiFePO4 > 10 A fuse > switch > +BATBUS > branches

One bus bar at the battery negative carries every return; the LED rail and the camera hub each have their own buck

